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Azure Data Lake Storage Gen2 Connector Guide

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Preface

Read the *Azure Data Lake Storage Gen2 Connector Guide* to learn how to set up and use Azure Data Lake Storage connections. Use a Azure Data Lake Storage Gen2 connection to securely write data to Microsoft Azure Data Lake Storage Gen2. Learn to perform create, append, and delete operations to manipulate data in the blob container.

CHAPTER 1

Introduction to Azure Data Lake Storage Gen2 Connector

Use Azure Data Lake Storage Gen2 Connector to securely perform create, append, and delete operations and write data to Microsoft Azure Data Lake Storage Gen2. Use Azure Data Lake Storage Gen2 Connector to write flat files with any content type, such as JSON and ORC.

Azure Data Lake Storage Gen2 Connector provides the following key features:

- Provides high performance and low latency data access.
- Supports a wide range of data types, including structured, semi-structured, and unstructured data.
- Stores raw data as a data lake before processing and loading it into a data warehouse.
- Provides a scalable storage solution for training machine learning models with large datasets.

Azure Data Lake Storage Gen2 overview

Azure Data Lake Storage Gen2 provides a scalable and secure data lake solution for high-performance analytics. It handles large volumes of data and integrates with Azure's analytics services.

Azure Data Lake Storage Gen2 provides the following key features:

- Stores and processes large volumes of data for big data analytics.
- Offers robust security features, such as encryption and integration with Azure Active Directory.
- Seamlessly integrates with various Azure services, such as Azure Databricks and Azure Synapse Analytics.

CHAPTER 2

Azure Data Lake Storage Gen2 connections

Use an Azure Data Lake Storage Gen2 connection to securely write data to an Azure blob container.

Azure Data Lake Storage Gen2 Connector supports a variety of entity types. You can also perform the following actions:

- Create a folder in a Data Lake Store account
- Append data to a file serially or concurrently
- Delete a file from a Data Lake Store account

After you create an Azure Data Lake Storage Gen2 connection, you can validate, test, and save the connection.

Note: Enable the **Blob soft delete** option for your storage event containers to prevent errors when creating or deleting files in your Azure storage account.

You can then publish the connection and click the **Metadata** tab to view the generated process objects for the connection.

Basic connection properties

The following table describes the basic properties that you can configure on the **Properties** tab of the connection creation page:

Property	Description
Name	Required. Unique name for the Azure Data Lake Storage Gen2 connection that identifies it in the Process Designer. The name must start with an alphabet and can contain only alphabets, numbers, and hyphens (-).
Location	The location of the project or folder where you want to save the connection. Click **Browse** to select a location. If the **Explore** page is currently active and a project or folder is selected, the default location for the connection is the selected project or folder. Otherwise, the default location is the location of the most recently saved asset.
Description	Optional. Description of the connection.

Property	Description
Type	Required. The type of connection you want to use for the connector or service connector. Select **Azure Data Lake Storage**.
Runtime Environment	Required. The runtime environment for the connection. You can run the connection on the Cloud Server.
Connection Test	Indicates whether the connection test was successful or not.
OData-Enabled	Not supported by Azure Data Lake Storage Gen2 Connector.
Authentication Type	Optional. Select your preferred authentication type and then configure the authentication-specific parameters. Select one of the following options: - Shared Key - Token Credentials
Account Key	This applies when **Authentication Type** is **Shared Key**. The access key for the Azure Data Lake Storage account.
Client ID	This property applies when you select the **Authentication Type** as **Token Credentials**. The client ID for your application registered in the Azure Active Directory.
Tenant ID	This property applies when you select the **Authentication Type** as **Token Credentials**. The tenant ID of the Azure Active Directory.
Client Secret	This property applies when you select the **Authentication Type** as **Token Credentials**. The client secret key generated for the client ID.

Azure Data Lake Storage Gen2 connection properties

To access the Azure Data Lake Storage instance, you need the account name and file system name.

The following table defines the Azure Data Lake Storage Gen2 connection properties that you must configure in the **Connection Properties** section:

Property	Description
Account Name	The name of the Azure Data Lake Storage account or service.
File System Name	The name of the Azure Data Lake Storage file system.

Azure Data Lake Storage Gen2 event targets

The event target properties allow you to control object creation and create an event service that can be called from Process Designer.

If you define an event target using the Azure Data Lake Writer, you can write content to the Azure Data Lake Storage Gen2.
All event targets also require the following properties:

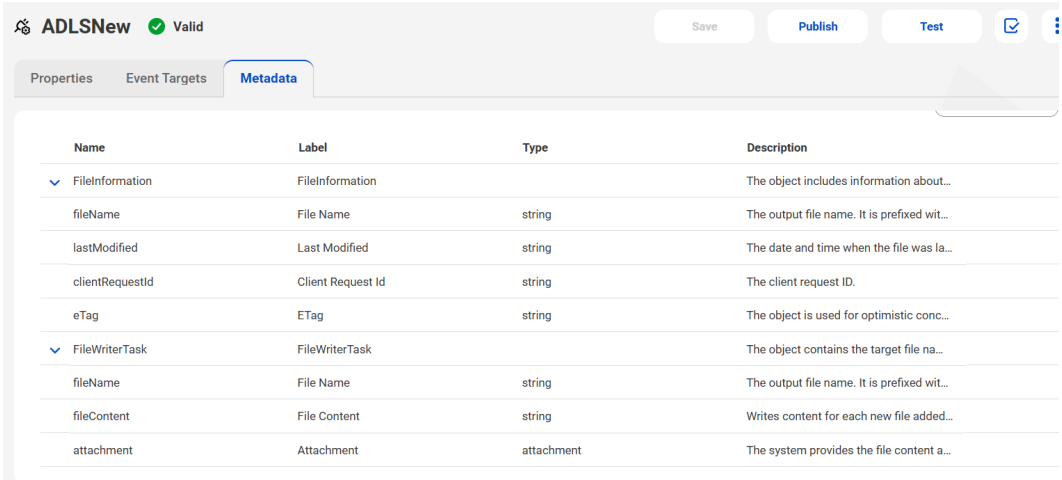
Amazon S3 Settings	Description
Name	Required. Specify a unique name of the target connection. If the provided name entered here does not exist, the event target automatically creates it using the Azure Data Lake Storage Gen2 settings specified in the connection properties. The question mark (?) character is not allowed in this field.
Operation	Required. The target operation. Select one of the following target operation from the list: - CREATE - APPEND - DELETE Default is **CREATE**.

Azure Data Lake Storage Gen2 connection metadata

After you create an Azure Data Lake Storage Gen2 connection, you can validate and save the connection. You can then publish the Azure Data Lake Storage Gen2 connection and click the **Metadata** tab to view the generated process objects for the connection.

When you publish an Azure Data Lake Storage Gen2 connection in Application Integration, the **Actions** and **Objects** appear on the **Metadata** tab.

The following image shows the **Metadata** tab for a published Azure Data Lake Storage Gen2 connection:



Azure Data Lake Storage Gen2 connection actions and objects

When you publish an Azure Data Lake Storage Gen2 connection in Application Integration, the **Actions** and **Objects** appear on the **Metadata** tab.

Use the **AzureDataLakeWriter** action to manage files in the target.

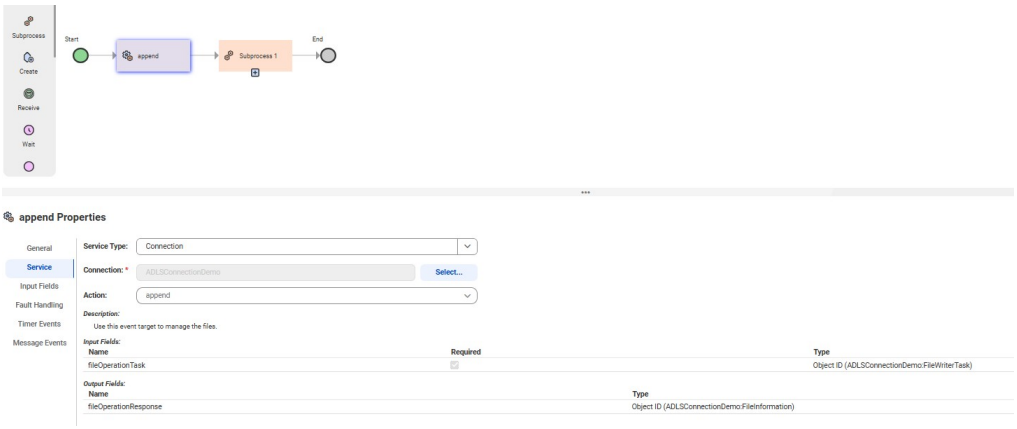
Consider the following information when you use specific objects:

Object Name	Description
FileInformation	The object includes information about the file but not its contents.
fileName	The output file name. It is prefixed with the relative file path.
lastModified	The date and time when the file was last modified.
clientRequestId	The client request ID.
eTag	The object is used for optimistic concurrency during updates.
FileWriterTask	The object contains the target file name and the string content to write to the target.
fileContent	Writes content for each new file added to the target file system.
attachment	Provides the file content as an attachment.

Using an Azure Data Lake Storage Gen2 connection in a process

After you create an Azure Data Lake Storage Gen2 connection, you can use it in a Service step in a process.

1. Add a Service step to the process.
2. Click the **Service** tab.
3. From the **Service Type** list, select **Connection**.
4. From the **Connection** list, browse and select the Azure Data Lake Storage Gen2 connection that you created.
5. From the **Action** list, select the action that you want to perform. For example, select the **append** action to append a file as shown in the following image:



In this case, Application Integration populates the **fileOperationTask** object under the input fields and **fileOperationResponse** object under the output fields.

6. Configure further steps as needed.
7. Validate, save, and publish the process.